Om Gawande

gawandeom2005@gmail.com | +91 8010173681 | linkedin.com/in/om-gawande-601081242 | github.com/omgawande1523

Summary

Artificial Intelligence undergraduate with practical experience in building real-world computer vision and NLP systems and deploying them on embedded and fully offline infrastructure. Experienced in training custom object detection models, developing local LLM-based agentic pipelines, and delivering end-to-end AI solutions from dataset preparation to production deployment. Interested in applied AI, computer vision, and scalable intelligent systems.

Experience

Research and Development Intern — Maharashtra Remote Sensing Application Centre (MRSAC) *Aug 2026*

- Contributed to a multi-hazard digital twin initiative under Maharashtra's climate resilience program, building a flood nowcasting module for Kolhapur city.
- Applied AI to a real-world scenario with incomplete and inconsistently available data, integrating satellite and geospatial datasets into a working nowcasting pipeline.

Data Analytics and AI Engineering Intern — Acube AI

- Developed AI-powered applications by integrating machine learning models, LLMs, and backend APIs into production-ready systems.
- Designed prompt engineering workflows, automated business processes, and collaborated on end-to-end AI solution deployment.

AI and ERP Solution Intern — RNR

- Developed AI modules for enterprise ERP applications, including machine learning pipelines, NLP workflows, and intelligent business automation features.
- Worked with cross-functional teams to design, implement, test, and deploy scalable AI solutions for real-world business use cases.

Skills

Programming Languages: Python, SQL
Deep Learning: CNNs, YOLO object detection, transfer learning, custom model training, TensorFlow, Keras
Computer Vision: OpenCV, image preprocessing, dataset annotation, real-time detection, camera integration
Generative AI: LLM applications, RAG pipelines, Ollama, vector databases (Qdrant), prompt engineering, agentic workflow design
Libraries and Tools: NumPy, Pandas, Matplotlib, Scikit-learn, Docker, Google Colab, Git, GitHub
Embedded AI and Hardware: Raspberry Pi 4, camera module integration, IoT
Soft Skills: Leadership, Problem Solving, Strategic Thinking, Teamwork, Adaptability

Projects

AI-Based Flood Nowcasting Digital Twin for Kolhapur — MRSAC

- Built a flood nowcasting module for Kolhapur city as part of a proposed multi-hazard (flood + landslide) digital twin architecture for the Upper Krishna sub-basin.
- Integrated open Sentinel-1/2 satellite imagery with MRSAC's LULC and DEM datasets, and developed an ML-based nowcasting layer with a dashboard for visualizing flood risk.

Local Area Network Agentic RAG System for Institutional Knowledge Base

- Built a LAN-based retrieval-augmented generation system using Ollama-hosted LLMs, Qdrant vector search, and MySQL, enabling secure on-premise deployment without external cloud services.
- Designed a multi-agent architecture (Router, SQL, RAG, Synthesis) with a dedicated verification agent for hallucination detection and answer grounding.
- Deployed via Docker Compose with a Django/DRF backend and React frontend for end-to-end query handling across the local network.

Smart AI Waste Segregation System using YOLO and Raspberry Pi

- Built an embedded YOLOv8-based computer vision system on Raspberry Pi for real-time autonomous waste classification and segregation.
- Handled dataset annotation, custom model training, and optimization for constrained edge hardware.

Sarcasm Detection for Low-Resource Languages

- Developing a transformer-based NLP framework for sarcasm detection in regional Indian languages using custom annotated datasets.
- Defined annotation guidelines, dataset schema, and strict separation of authentic and synthetic data for research integrity.

Education

B.Tech in Artificial Intelligence (2027) *CGPA: 8.68 / 10*

St. Vincent Pallotti College of Engineering and Technology, Nagpur

Certifications & Achievements

- President, AI Prabodha Student Forum (previously Secretary & Founding Member)
- Class Representative (2023–Present)
- Fundamentals of Deep Learning — NVIDIA
- Introduction to Machine Learning — NPTEL
- Data Engineering in Cloud Environment Training
- Generative AI & DevOps Faculty Development Program